

Composing Unseen Behavior Combinations Across Worlds with Transplantable Dynamics

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Abstract—World models can predict how environments evolve, but their learned dynamics are rarely exposed as editable components. We ask whether a world model can instead expose its dynamics through an intervenable latent interface, so that what it encodes about the dynamics can change without changing the predictor itself. We factorise the context of a conditional predictor into dynamical factors (stride, boundary, action rotation) and test whether these factors can be transplanted across environments and selectively rewritten after a local dynamics-rule change. Under combination-ODD, factors inferred solely from donor rollouts transfer to held-out joints, achieving 0.75+/-0.05 prediction accuracy and 0.78+/-0.04 reachable-goal planning success without target-world transitions. We then flip the boundary rule mid-episode. Rewriting only the corresponding 16-dimensional factor from a shared post-change observation buffer yields planning-success AUC 20.58 (labelled edge-prototype oracle 21.25), outperforming broader and matched-budget adaptations with the shared predictor frozen. Matched wrong-factor and random 16-dimensional controls show that the benefit is localised to the affected factor. These results provide evidence that factorised context can function not merely as a representation, but as an editable interface to learned dynamics.

Index Terms—world models, intervenable representations, compositional generalization, dynamics adaptation, model-based reinforcement learning

I. INTRODUCTION

*Can you change what a world model believes about the physics
without changing the world model itself?*

World models typically expose learned dynamics through prediction and rollout, rather than as quantities that can themselves be directly manipulated. We ask whether learned dynamics can instead become something to *operate on*: if a dynamics factor is represented by a local component of a model’s context, can that component be transplanted across environments or rewritten when the corresponding factor changes?

Prior work supplies several ingredients. Joint-embedding predictive architectures learn representations by predicting in embedding space [1]; world models condition shared predictors for control [2]; skill-embedding and meta-RL methods encode transferable context for shared policies [3], [4]; compositional and object-centric models factorise representations to improve generalisation [5]; disentanglement seeks components aligned with factors of variation [6]. These lines of work show that shared predictors can be conditioned on structured context,

but they do not establish that such context is an *operational interface* to learned dynamics—i.e., that editing the factor tied to a changed dynamics rule recovers planning more effectively than matched wrong-factor or random edits under the same observations.

We therefore keep the predictive substrate fixed and ask a sharper question: does factorisation make dynamical context *intervenable*? We use a conditional predictor $\text{Pred}(z, a, m)$ whose context $m = (m_s, m_e, m_r)$ represents stride, boundary, and action rotation. The same frozen Pred is evaluated under assembled priors

$$m_s^A \oplus m_e^B \oplus m_r^C \xrightarrow{\text{Pred}} \text{PLAN}.$$

A factor inferred in one environment can be combined with factors from others, or rewritten after the corresponding dynamics rule changes. Downstream planning is the readout: interventions must change not only the latent, but the behaviour induced by the learned dynamics.

The experiments remove alternative explanations in turn. Labelled slot assembly establishes a capacity ceiling—whether Pred can execute a correctly supplied dynamical prior—without claiming that factors are recoverable from experience. Transplantation tests whether individual factors transfer across held-out joints while preserving the rest. Inference removes access to target-world transitions and asks whether factors can instead be recovered from donor rollouts. A mid-episode change to one dynamics rule then tests selective rewriting: whether editing only the corresponding factor recovers the new dynamics.

These interventions also test *intervention locality*. If rewriting the factor tied to the changed rule improves planning, while matched wrong-factor and random edits—and broader full-factor, monolithic, or full-predictor adaptation under the same observations—do not match that recovery, the result provides evidence that the factor carries localised dynamical information rather than merely providing arbitrary latent capacity.

a) *Claim.*: A factorised latent context can provide a controllable interface to learned dynamics, evidenced by empirical intervention locality under planning. We do not claim unique identification of physical constants, and we do not propose a new world-model architecture; we study whether a standard conditional world model can expose dynamics in a form that can be transplanted, selectively rewritten, and evaluated through downstream control.

b) Contributions.: (1) An intervention protocol for learned dynamics: composition and specificity under transplantation, inferred transfer into planning, and selective editing after a dynamics-rule change. (2) Inferred transplants support reachable-goal planning below the labelled ceiling (0.75 ± 0.05 prediction accuracy; 0.78 ± 0.04 planning success) without target-world transitions. (3) Selective rewriting after a boundary-rule change: the correct 16-dimensional edit outperforms matched wrong/random controls (locality) and matched-budget broader adapters; the gap vs. broader adapters is largest on stale-fail episodes, and locality persists under low observation budgets.

Code and experiment configurations will be released upon acceptance.

II. METHOD

We study an editable dynamics interface built on a shared conditional world model, keeping the predictive substrate fixed while intervening on its environment-dependent context.

A. Predictive substrate and domain

a) Transition systems.: Let W be a family of deterministic environments over states $x \in \mathcal{X}$ and actions $a \in \mathcal{A}$. Each world $w \in W$ induces a transition kernel $T_w(x'|x, a)$. In our primary domain, $\mathcal{X}$ is a 12×12 grid and $\mathcal{A}$ consists of the four cardinal moves. Worlds differ by three discrete dynamics rules: stride $s \in \{1, \dots, 7\}$, boundary $e \in \{\text{clamp}, \text{wrap}\}$, and action rotation $r \in \{\text{none}, \text{cw}, \text{ccw}\}$. A world is therefore a joint $w = (s, e, r)$. This controlled setting is chosen so that factor-local interventions admit exact matched controls and stale-fail diagnostics; we do not claim it substitutes for continuous control benchmarks.

b) Conditional world model.: A shared encoder, predictor, and decoder implement

$$z = \text{Enc}(x), \quad z' = \text{Pred}(z, a, m), \quad \hat{x}' = \text{Dec}(z'). \quad (1)$$

The context m carries environment-dependent dynamics, while $\text{Enc}/\text{Pred}/\text{Dec}$ are shared across W and remain frozen during the interventions below. We factorise the context as

$$m = \text{concat}(m_s, m_e, m_r) \in \mathbb{R}^{32+16+16}, \quad (2)$$

where the three blocks are treated as candidate local handles for stride, boundary, and action rotation, respectively. Equation (1) defines the frozen predictive substrate on which we intervene; the factorisation itself is evaluated empirically by the experiments below.

c) Combination-OOD split.: We partition $W = W_{\text{train}} \cup W_{\text{hold}}$ with $|W_{\text{train}}| = |W_{\text{hold}}| = 21$. Every individual factor value appearing in a held-out joint also appears in at least one training world, but the held-out joint itself does not. Thus, the evaluation requires recombination of known dynamics-factor values rather than extrapolation to unseen factor values. Checkpoints are selected using W_{train} only; held-out transition kernels T_C for $C \in W_{\text{hold}}$ are never used as training or checkpoint-selection targets.

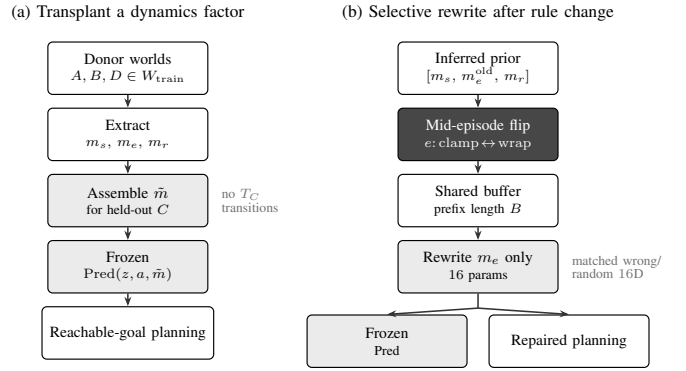


Fig. 1. Editable dynamics interface. (a) Transplant factor blocks across worlds with Pred frozen. (b) After a mid-episode boundary flip, rewrite only m_e from a shared buffer.

B. Interfaces to m

a) Labelled slots.: For each factor value, we learn a corresponding embedding. A labelled prior is assembled by concatenating the appropriate factor embeddings as in (2). This interface defines a capacity ceiling: it tests whether the frozen Pred can execute a correctly supplied dynamical prior, rather than whether the factors can be recovered from experience.

b) Inferred factors.: To remove access to factor labels at test time, a permutation-invariant M -encoder maps a bag of K observed transitions to $(\hat{m}_s, \hat{m}_e, \hat{m}_r)$ without receiving factor indices as input. On training worlds only, we jointly train $\text{Enc}/\text{Pred}/\text{Dec}$ and the M -encoder. Let $z = \text{Enc}(x)$, $z' = \text{Enc}(x')$, and $\hat{z}' = \text{Pred}(z, a, m)$ for a context m assembled from the M -encoder. The predictive objective combines latent regression with reconstruction,

$$\begin{aligned} \mathcal{L}_{\text{pred}} = & \|\hat{z}' - z'\|_2^2 + \lambda_{\cos}(1 - \cos(\hat{z}', z')) \\ & + \lambda_{\text{ncc}} \text{InfoNCE}(\hat{z}', z') \\ & + \lambda_{\text{vr}}(\text{VICReg}(z) + \text{VICReg}(\hat{z}')) \\ & + \lambda_{\text{dec}}(\|\text{Dec}(z) - x\|_2^2 + \|\text{Dec}(\hat{z}') - x'\|_2^2). \end{aligned} \quad (3)$$

For factorised inference we add three train-only terms on the inferred blocks. *Consistency* is a supervised contrastive loss $\mathcal{L}_{\text{con}}$ that pulls together blocks whose factor value matches and repels mismatches (temperature $\tau=0.1$). *Invariance* is a within-value variance penalty

$$\mathcal{L}_{\text{inv}} = \sum_{f \in \{s, e, r\}} \mathbb{E}_v [\|\hat{m}_f - \bar{m}_f(v)\|_2^2]$$

over train worlds sharing factor value v . *Multi-view* draws a second independent context from the same world and applies $\mathcal{L}_{\text{view}} = \sum_f \|\hat{m}_f - \hat{m}_f^{\text{view}}\|_2^2$. Recombination (“swap”) further trains Pred on concatenated blocks drawn from different train worlds that share the required factor values, so the predictor sees off-joint assemblies during training. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}} + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}} + \lambda_{\text{view}} \mathcal{L}_{\text{view}} \quad (4)$$

with hyperparameters fixed a priori (reported checkpoints use $\lambda_{\text{con}}=0.5$, $\lambda_{\text{inv}}=1$, $\lambda_{\text{view}}=1$, swap probability 1). At test

time, operational results use inferred factors unless a protocol explicitly specifies labelled slots.

C. Intervention I: transplanting a dynamics factor

Fix a held-out target world $C = (s_C, e_C, r_C) \in W_{\text{hold}}$. Donor worlds $A, B, D \in W_{\text{train}}$ supply the required factor values. Transplantation constructs a prior $\tilde{m}$ for C and evaluates whether the frozen predictor behaves according to the target transition system T_C (Figure 1, left).

a) *Labelled transplant.*: We first construct $\tilde{m} = \text{concat}(m_{s_C}, m_{e_C}, m_{r_C})$ using labelled slots from training worlds. All valid donor triples produce the same $\tilde{m}$ by construction (unique $m = 1$); therefore, donor invariance in this setting is architectural rather than an empirical discovery. The empirical test is whether the resulting frozen predictor agrees with T_C , and whether performance collapses when one required factor is replaced by a mismatched donor.

b) *Inferred transplant.*: We then infer each factor from a short rollout of a donor exhibiting the corresponding dynamics rule: $\hat{m}_s$ from a stride- s_C donor, $\hat{m}_e$ from an edge- e_C donor, and $\hat{m}_r$ from a rotation- r_C donor. The inferred blocks are concatenated to form the transplanted prior. Because inferred vectors may vary with the donor, we report both prediction accuracy against T_C and agreement across donor choices. No transitions from the target world C are observed during prior construction. Wrong-factor donors (correct slot index, incorrect law) are the negative control for intervention locality.

c) *Planning readout.*: For a transplanted prior $\tilde{m}$, we construct a one-step transition graph using $\text{Enc} \rightarrow \text{Pred}(\cdot, \cdot, \tilde{m}) \rightarrow \text{Dec}$, snapped to the grid, and run breadth-first search (BFS) to a goal. Ground-truth transitions are used only to execute the selected action and to filter reachable goals; they are never supplied as transitions inside the planner. Success is defined as occupying a reachable goal within a fixed horizon (reachable-goal evaluation).

D. Intervention II: selective rewriting after a dynamics change

Transplantation edits dynamics *across* worlds. Selective rewriting instead edits dynamics *within* an episode after the simulator changes one dynamics rule (Figure 1, right).

a) *Protocol.*: We initialise each episode from an inferred transplanted prior for the pre-change world. At a fixed mid-episode step, exactly one simulator factor is changed. Reported rewrite, locality, and adaptation results use the boundary flip $e : \text{clamp} \leftrightarrow \text{wrap}$; other single-factor flips are supported by the protocol but are not claimed as evidenced here. A post-change observation buffer containing 24 random-walk transitions is generated once and frozen. Every adaptation method receives the same prefix of length $B \in \{1, 2, 4, 8, 12, 24\}$.

b) *Factor-local rewrite.*: For the boundary intervention, only the boundary block is replaced:

$$[m_s, m_e^{\text{old}}, m_r] \longrightarrow [m_s, m_e^{\text{new}}, m_r].$$

The new m_e is obtained from the M -encoder applied to the shared post-change prefix. $\text{Enc}/\text{Pred}/\text{Dec}$ remain frozen. This intervention tests whether the learned representation of the changed law can be updated through a single local handle.

c) *Controls.*: We compare selective rewriting against the following controls, all using the same post-change observation buffer: *Full-factor*: re-infer all three context blocks from the same prefix. *Monolithic*: optimise the unstructured concatenated m by SGD, updating all 64 context parameters. *Full-predictor*: fine-tune Pred for a fixed number of epochs on the same buffer. *Oracle*: replace m_e with a training-world prototype corresponding to the true new boundary law; this uses label access but no target-world transitions. *Matched locality controls*: update the wrong block m_r (16 parameters), or update a random 16-dimensional subspace of m . Hyperparameters are fixed a priori and are not tuned on held-out episodes.

E. What constitutes evidence for intervention locality

We do not claim that m_s, m_e, m_r are uniquely identified physical constants. Instead, we test for *intervention locality* under planning. Specifically, we ask whether:

- 1) transplanting the intended factor improves agreement with the target transition system while mismatched factors cause a substantial degradation; and
- 2) after a local dynamics change, rewriting the corresponding block recovers planning more effectively than an equally sized wrong or random intervention, as well as broader unstructured or full-model adaptation given the same post-change observations.

Prediction accuracy, donor agreement, reachable-goal success, and the area under the planning-success-versus- B curve provide the corresponding readouts.

III. EXPERIMENTS

We evaluate whether factorised dynamical context can serve as an editable interface to environment dynamics. We proceed from composition and specificity, through inference and planning without target-world transitions, to the central test: selective rewriting after a mid-episode dynamics change, compared against matched-dimensional and broader adaptation under identical post-change observations.

A. Experimental setup

a) *Domain.*: As in Section II-A: 12×12 grid; stride, boundary, and action-rotation dynamics rules; 21/21 combination-OOD train/hold. Held-out kernels are excluded from training and checkpoint selection.

b) *Readouts and planning.*: We report one-step Pred accuracy against T_C as a diagnostic of factor correctness, donor agreement, and reachable-goal planning success. One-step Pred is intentionally local: it attributes errors to individual factors without compounding. The primary claim, however, is evaluated under multi-step control—given m , the planner builds an imagined transition graph by applying $\text{Enc} \rightarrow \text{Pred} \rightarrow \text{Dec}$ at every grid cell and action, then runs BFS—so planning success requires consistent imagined dynamics along entire trajectories, not a single transition. Ground-truth transitions are used only for action execution and reachable-goal filtering, never for prior construction or planner

expansion. Absolute planning rates are therefore reachable-goal rates, not open-world rates; labelled slots are a capacity ceiling only. Labelled Pred of 0.933 (rather than 1.0) reflects residual error in the shared predictive substrate under discrete snap-to-grid decoding, even with correctly assembled slots.

c) *Adaptation metric.*: For selective rewriting we vary the shared post-change buffer length $B \in \{1, 2, 4, 8, 12, 24\}$ and plot mean reachable-goal success against B . We summarise each curve by its trapezoidal area (AUC):

$$\text{AUC} = \sum_{i=1}^5 \frac{1}{2} (s(B_i) + s(B_{i+1})) (B_{i+1} - B_i),$$

where $s(B)$ is mean success at budget B . A perfect curve ($s \equiv 1$) yields $\text{AUC} = 23$. We also report mean success over $B \in \{1, 2, 4\}$. Hyperparameters were fixed a priori and not tuned on held-out episodes.

B. Factor composition and specificity

a) *Learned dynamics factors compose across held-out joints.*: For each held-out target C , we assemble $m = \text{concat}(m_s^A, m_e^B, m_r^D)$ from labelled train-world slots matching C 's factor values, with Pred frozen. Donor enumeration yields 4,620 assemblies with unique $m = 1$ by construction (architectural donor invariance). Mean Pred accuracy vs. T_C over the 21 held-out worlds is 0.933 (Table I). On a separate reachable-goal planning suite ($n=101$ episodes), the same labelled-slot prior yields success ≈ 0.90 . A frozen predictor can decode dynamics from factor combinations never jointly observed in training—compositional capacity, not factor recoverability.

b) *Composition is factor-specific.*: Replacing the stride block by a mismatched donor collapses Pred accuracy from 0.933 to ≈ 0.09 ; wrong-edge donors degrade Pred less severely (≈ 0.66), so specificity is factor-dependent rather than uniformly catastrophic. This asymmetry is expected geometrically: on a finite grid, the boundary rule clamp vs. wrap changes the next state only for actions that would cross the border, whereas stride affects every transition. One-step Pred accuracy therefore understates boundary errors relative to stride errors; we treat wrong-edge Pred as a weak negative control and rely on planning after an explicit mid-episode boundary flip (Section III-D) for the primary locality claim. Successful composition depends on factor identity, not merely a useful 64-dimensional context. These tests establish distinguishable dynamical roles for the blocks; they do not claim unique identification of physical constants.

C. Inferred transplantation and planning

a) *Factors transfer without target-world transitions.*: We infer $(\hat{m}_s, \hat{m}_e, \hat{m}_r)$ from donor rollouts and transplant into the held-out joint. Inference yields 0.75 ± 0.05 Pred accuracy over three seeds (donor agree ≈ 0.78), below the labelled ceiling 0.933 but far above the wrong-stride collapse ≈ 0.09 . Useful dynamical information can be recovered from donor experience and transferred to a held-out joint—without implying perfect factor recovery.

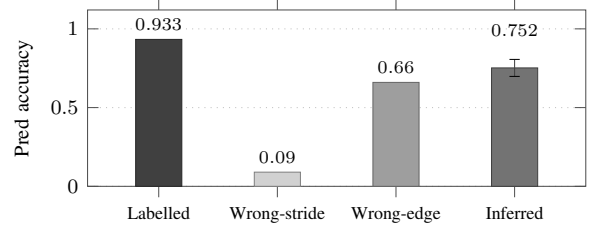


Fig. 2. Composition and specificity (Pred accuracy). Labelled slots compose on held-out joints; wrong-stride donors collapse Pred; wrong-edge donors degrade less. Inferred transplant is mean $\pm$ std over three seeds.

TABLE I

TRANSPLANTATION ON HELD-OUT JOINTS. LABELLED PRED IS THE MEAN OVER 21 HELD-OUT WORLDS (4,620 ASSEMBLIES SHARE ONE m BY CONSTRUCTION); LABELLED PLANNING IS A SEPARATE SUITE ($n=101$). WRONG-STRIDE IS THE STRONG NEGATIVE CONTROL; INFERRED TRANSPLANTATION USES DONOR EXPERIENCE WITHOUT TARGET-WORLD TRANSITIONS.

Transplant setting	Pred accuracy $\uparrow$	Planning success $\uparrow$
Labelled slots	0.933	≈ 0.90
Wrong-stride donors	≈ 0.09	—
Inferred transplant	0.75 ± 0.05	0.78 ± 0.04

b) *Transferred factors support planning.*: Under the same integrity constraints, donor-inferred contexts achieve 0.78 ± 0.04 reachable-goal success vs. ≈ 0.90 for the labelled-slot ceiling on the same planning suite. The factors are therefore operationally useful for action under imagined dynamics, not only for one-step prediction. Composition, specificity, inference, and planning establish the prerequisites for an editable interface; we next ask whether it can be surgically edited when dynamics change.

D. Selective rewrite and locality controls

a) *A dynamics change can be repaired by editing one factor.*: Primary evidence is a mid-episode boundary-rule change (e : clamp $\leftrightarrow$ wrap). On 315 held-out episodes we initialise from an inferred pre-change prior, freeze one shared post-change buffer of 24 transitions, and give every method the same prefix of length B . Selective rewriting replaces only the edge block,

$$[m_s, m_e^{\text{old}}, m_r] \longrightarrow [m_s, m_e^{\text{new}}, m_r],$$

with m_e^{new} from the M -encoder on the shared prefix; Enc/Pred/Dec stay frozen. AUC is 20.58 vs. oracle edge prototype 21.25 (Table II, Figure 3): most oracle-level planning is recovered by editing only the context block associated with the changed dynamics rule.

b) *The benefit is localised to the affected factor.*: Matched controls update wrong block m_r or a random 16-dimensional subspace under the same data and budget, yielding AUC 16.11 and 16.50. This provides evidence for intervention locality relative to the matched controls tested here—not that no alternative 16-dimensional subspace could encode the change.

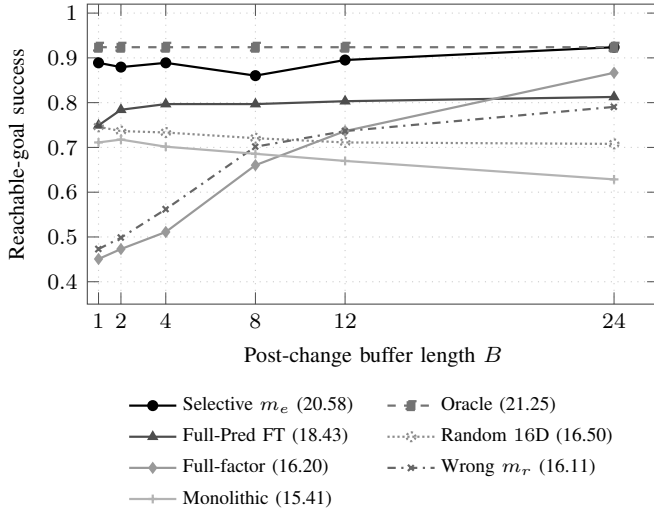


Fig. 3. Selective editing under shared post-change buffers ($n=315$). AUC is trapezoidal area under mean reachable-goal success vs. B (maximum 23; parentheticals in the legend). Selective m_e tracks the labelled edge-prototype oracle and beats matched 16-dim and broader adapters.

TABLE II

SELECTIVE EDITING AFTER MID-EPISEODE BOUNDARY-RULE CHANGE ON 315 EPISODES (IDENTICAL PREFIXES). MATCHED 16-DIM CONTROLS TEST LOCALITY; BROADER ADAPTERS SHARE THE SAME BUFFER.

Adaptation	Updated params	AUC $\uparrow$
Oracle (edge prototype)	—	21.25
Selective rewrite (m_e)	16	20.58
Full-predictor FT	all Pred	18.43
Random 16-dim subspace	16	16.50
Full-factor re-inference	64	16.20
Wrong factor (m_r)	16	16.11
Monolithic prior SGD	64	15.41

AUC definition. Trapezoidal area under mean reachable-goal success $s(B)$ vs. buffer length $B \in \{1, 2, 4, 8, 12, 24\}$: $\sum_i \frac{1}{2}(s(B_i) + s(B_{i+1}))(B_{i+1} - B_i)$. Perfect curve $s \equiv 1$ gives AUC = 23. See also Section III-A.

E. Adaptation baselines and stale-fail episodes

a) Targeted editing outperforms broader adaptation under matched data. Full-factor re-inference (16.20), monolithic SGD on m (15.41), and full-predictor fine-tuning (18.43) all trail selective rewrite despite more trainable parameters and the same observations. These results suggest that the advantage is structural: adapting more broadly does not reproduce the benefit of editing the factor tied to the changed dynamics rule.

b) Targeted editing repairs episodes that stale dynamics cannot solve. Before comparing adapters, we fix the $n=69$ episodes where planning already fails under the unchanged pre-change prior. On this subset, selective rewrite AUC is 13.84 vs. 3.52 (full Pred) and 2.78 (monolithic); oracle is 16.00 (Table III). The gap is largest where adaptation is genuinely necessary.

TABLE III

STALE-FAIL DIAGNOSTIC ($n=69$): PLANNING FAILS UNDER THE STALE PRIOR; SUBSET FIXED BEFORE ADAPTER COMPARISON.

Method	Stale-fail AUC $\uparrow$
Oracle	16.00
Selective rewrite (m_e)	13.84
Full-predictor FT	3.52
Monolithic prior SGD	2.78

AUC definition. Same trapezoidal AUC as Table II, restricted to the $n=69$ stale-fail episodes (perfect curve still = 23).

F. Low-data adaptation

a) Selective editing is effective from very small post-change buffers. Averaged over $B \in \{1, 2, 4\}$, selective rewrite succeeds at 0.89 vs. 0.51 (wrong 16-dim) and 0.74 (random 16-dim). At $B=1$, selective rewriting already exceeds 90% of the oracle’s final reachable-goal success. Routing adaptation to the factor associated with the changed dynamics rule remains effective when post-change experience is scarce.

b) Takeaway. Labelled transplantation establishes compositional capacity; wrong-factor transplantation shows specificity; inferred transplantation transfers into planning without target-world transitions; selective rewriting after a boundary-rule change recovers near-oracle planning with Pred frozen. Matched wrong/random controls support locality; broader adapters and the stale-fail diagnostic show that the recovery is consequential rather than cosmetic; low- B evaluation shows it remains useful under scarce observations.

IV. LIMITATIONS

Our conclusions are deliberately scoped to the intervention regime studied here. The experiments use a controlled deterministic gridworld in which the relevant dynamics factors are known and discretely parameterised. We adopt this domain not as a substitute for DeepMind Control or visual world models, but because it makes intervention locality *testable*: matched-dimensional edits, wrong-factor donors, and stale-fail filtering require known factor axes and exact transition kernels. Whether the same locality emerges in continuous physics, high-dimensional visual environments, contact-rich dynamics, or physical robots remains to be established. The factorisation $m = (m_s, m_e, m_r)$ is likewise architecturally imposed (fixed block sizes), not discovered by the model; our claim is that such blocks can function as an editable interface, not that the world model invents the factorisation. Our evidence for factor semantics is empirical rather than a formal identifiability result: the correct-factor interventions outperform matched wrong and random controls, but this does not establish unique correspondence between latent blocks and dynamics factors. Finally, the primary adaptation experiment considers a single discrete rule change (boundary clamp $\leftrightarrow$ wrap) while keeping the predictive substrate fixed, under reachable-goal evaluation. Simultaneous or gradual changes, and changes outside the

learned factor vocabulary, may require richer representations or adaptation mechanisms.

These boundaries specify where the demonstrated intervention principle has yet to be tested, rather than altering the central result: within the controlled setting, targeted modification of the factor associated with a changed dynamics rule consistently outperforms unrelated or broader interventions under matched observations.

V. CONCLUSION

We studied whether learned dynamics can be exposed as an intervenable representation rather than only as a quantity to predict. A conditional world model with factorised dynamical context composes across held-out joints, preserves factor-specific behaviour under transplantation, and supports inference from donor experience without target-world transitions. More importantly, after a mid-episode change to one dynamics rule, rewriting only the corresponding context block recovers planning performance close to an oracle while the shared predictive substrate remains frozen.

The resulting picture is not that factorisation automatically produces an identifiable world model. Rather, it suggests a more operational criterion for learned dynamics: a useful factor should support controlled intervention under planning, outperform matched wrong or random edits, and produce consequences when the corresponding dynamics rule changes. Under this criterion, the learned context becomes an interface that can be transplanted across environments and selectively rewritten when dynamics change.

We hope this perspective shifts evaluation of world models from asking only whether they predict the future to also asking whether their learned representations can be *operated on*. Establishing such intervention locality in richer physical environments is a natural next step toward world models whose internal dynamics can be inspected, transferred, and adapted without retraining the entire model.

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Acknowledgments appear in the camera-ready version.

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APPENDIX

a) Integrity.: Search never queries ground-truth transitions; GT is restricted to execution, reachable-goal filtering, and labelled oracle ceilings.

b) Matched controls.: An earlier wrong-slot baseline edited more than 16 parameters; the matched 16-dim suite is the primary fairness control for intervention locality.

c) Hold-self.: System identification from the target world’s own rollouts reaches approximately 0.81 for factorised and monolithic encoders; we report it only as contrast, not as cross-world editing.